

ML Summer Bridge: Week 0 Study Notes

Machine Learning Class

Summer 2026

Week 0: Getting Started with Google Colab

1. What is Google Colab?

Google Colab is a free, browser-based tool that allows you to write and execute Python code. It is specifically designed for Data Science and Machine Learning because it comes pre-installed with powerful libraries like NumPy, Matplotlib, and Pandas.

2. How to Set Up Sign In: Go to colab.research.google.com and sign in with your school Google account. Create a Notebook: Click "New Notebook" at the bottom of the pop-up.

The Interface: * Code Cells: Where you write Python. Press Shift + Enter to run a cell.

Text Cells: Where you write notes (like this guide).

Importing Libraries: To use NumPy, you must "import" it at the very top of your code. We must do the same thing if we want to use Pandas. Type this into the first cell and run it:

```
import numpy as np
import pandas as pd
print(np.__version__)
print(pd.__version__)
```

Note: We use 'as np' and 'as pd' so we can type 'np' and 'pd' instead of the full words 'numpy' and 'pandas' every time.

When you run the short code segment above (by clicking the little arrow next to the cell), Colab should display the versions of numpy and pandas that are installed. As of April 26 these are 2.0.2 and 2.2.2 respectively, but are subject to change. As long as you get a number and the code runs you are in good shape.

From this point we will take a temporary detour to work on our basic Python skills. For those of you who have taken design engineering in 10th grade and therefore have experience with Java you will notice many similarities between Python and Java. We will be using CodeHS to refresh our Python skills. The key things we want to brush up on are similar to the topics we covered in Java in 10th grade design engineering. These include: variables, conditionals (if and if-else), loops and functions. If you are already familiar with these Python topics you can skip this first of the Python review. However, if you are a little rusty in your Python I highly recommend doing this section, as the later sections build upon this.

The name of the CodeHS course we will be following is 'Intro to Python Programming.' The relevant units we will cover are units 3, 5, 7 and 9. Unit 3 covers variables, print statements and a few other basic operators. Unit 5 covers conditionals. Unit 7 covers loops and unit 9 covers functions. It should only take you a few hours to complete all four units.

Once you have brushed up on your Python, you can return to Google Colab. During the summer bridge we will consider a few running examples. We will introduce the first example here.

Father's Height	Mother's Height	Gender	Height
70	60	M	63
68	62	F	60
73	64	M	67
72	64	F	61
74	68	F	69

We will analyze some aspects of this table next week.